

Data Cleaning and Visualization of Technological Products Dataset Using Python

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Introduction

Data cleaning and visualization are essential steps in the data science workflow. Raw datasets often contain inconsistencies, missing values, duplicate records, and outliers that can affect the accuracy of analysis. The objective of this project is to clean, process, and analyze a technological products dataset and generate meaningful insights through data visualization.

The project was implemented using Python and popular data science libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

Objective

The main objectives of this project are:

- To understand the structure of the dataset.
- To perform data cleaning and preprocessing.
- To identify missing values, duplicate records, and outliers.
- To visualize important patterns and trends in the dataset.
- To generate meaningful insights through exploratory data analysis.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Dataset Description

The dataset contains information about various technological products from different brands. The attributes included in the dataset are:

- Brand
- Device
- Model
- Country of Origin
- Date of Release
- Price (USD)

The dataset consists of products from major technology companies such as Apple, Samsung, Google, Sony, Microsoft, Dell, HP, Lenovo, Asus, and Acer.

Data Cleaning Process

The following preprocessing steps were performed:

1. Dataset Inspection

The dataset structure was analyzed using functions such as:

- shape()
- columns()
- info()
- describe()

2. Removal of Unnecessary Columns

An unwanted column named "Unnamed: 0" was removed from the dataset to improve data quality. **3. Missing**

Value Analysis

The dataset was checked for missing values using the isnull() function.

Result: No missing values were found in the dataset.

4. Duplicate Record Analysis

Duplicate records were identified using the duplicated() function.

Result: No duplicate records were found.

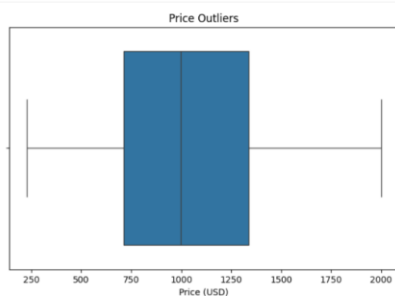
5. Data Transformation

The Date of Release column was converted into datetime format, and a new Release Year column was extracted for analysis.

6. Outlier Detection

A boxplot was created to identify outliers in product prices. A few premium products were observed to have significantly higher prices compared to others.

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Price (USD)'])
plt.title("Price Outliers")
plt.show()
```



Data Visualization and Analysis

Visualization 1: Products by Brand

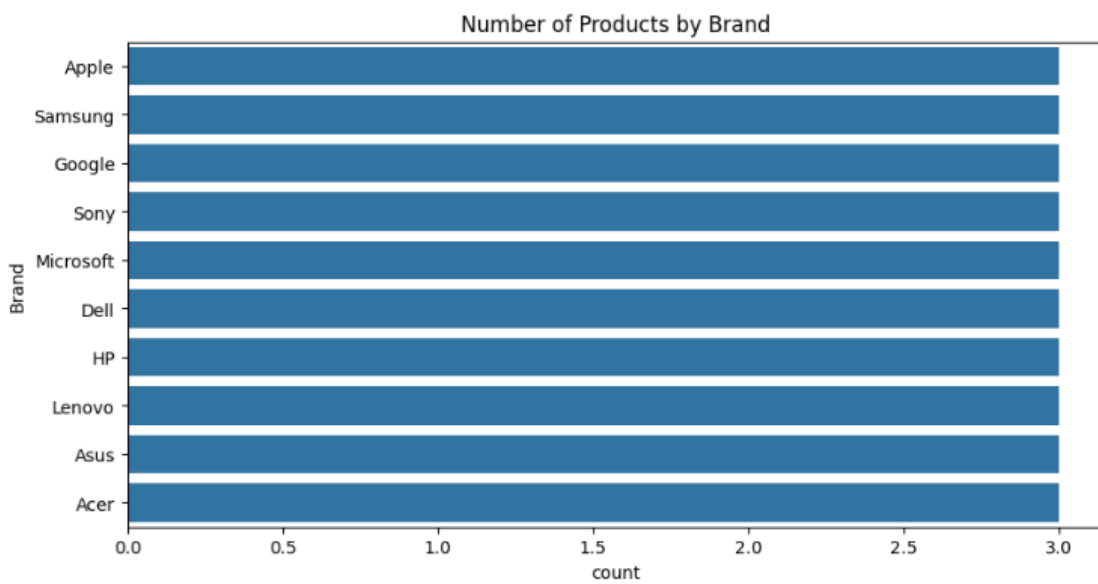
A count plot was created to analyze the number of products available for each brand.

Observation: Each brand contributes three products to the dataset, resulting in a balanced distribution across all brands.

```
plt.figure(figsize=(10,5))

sns.countplot(
    y='Brand',
    data=df,
    order=df['Brand'].value_counts().index
)

plt.title("Number of Products by Brand")
plt.show()
```



Visualization 2: Products by Country

A country-wise product distribution chart was generated.

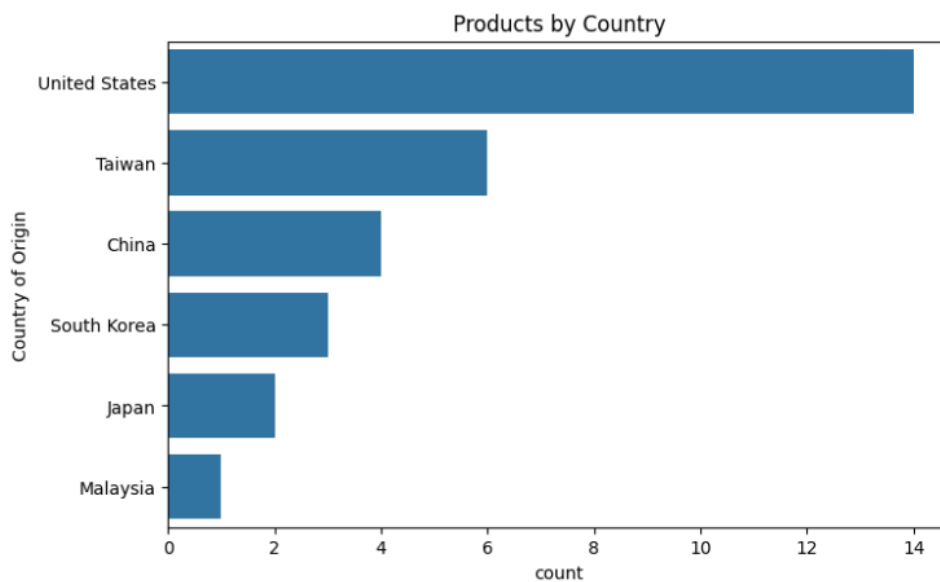
Observation: The United States contributes the highest number of products, followed by Taiwan and China.

Interpretation: This reflects the strong presence of technology companies headquartered in the United States.

```
plt.figure(figsize=(8,5))

sns.countplot(
    y='Country of Origin',
    data=df,
    order=df['Country of Origin'].value_counts().index
)

plt.title("Products by Country")
plt.show()
```



Visualization 3: Average Product Price by Brand

A bar chart was created to compare the average product prices across different brands.

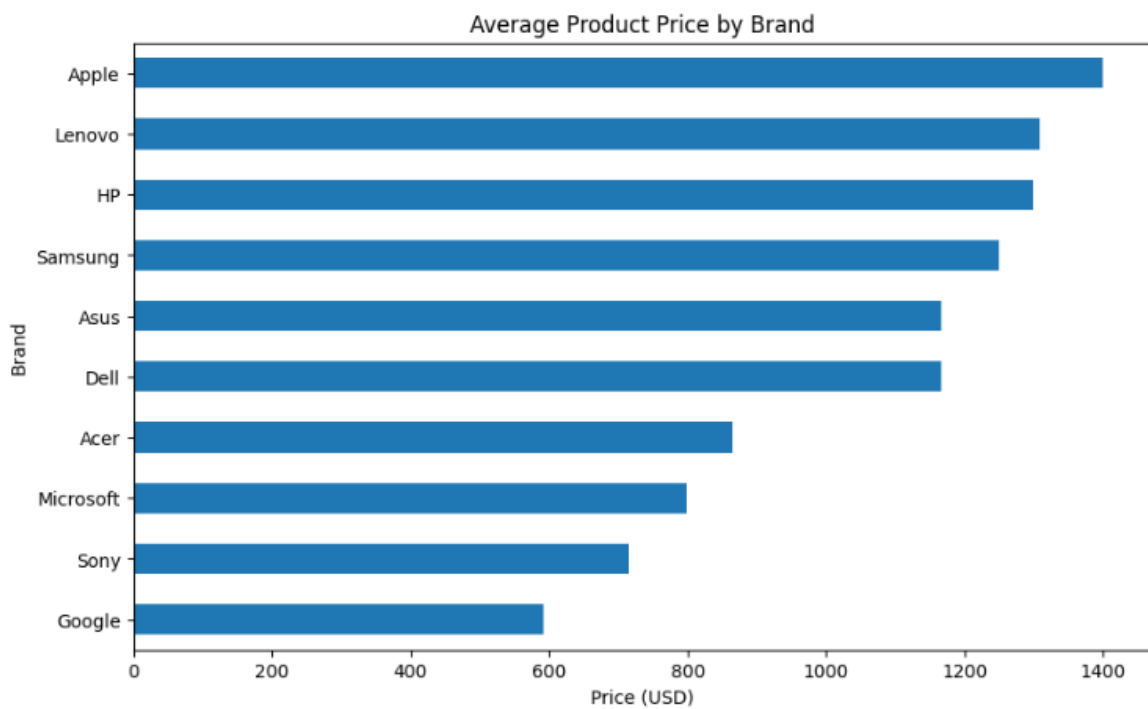
Observation: Apple has the highest average product price, while Google has the lowest average product price.

Interpretation: The results indicate different market positioning strategies among brands, ranging from premium to budget-friendly products.

```
brand_price = df.groupby('Brand')['Price (USD)'].mean().sort_values()

brand_price.plot(
    kind='barh',
    figsize=(10,6)
)

plt.title("Average Product Price by Brand")
plt.xlabel("Price (USD)")
plt.show()
```



Visualization 4: Price Distribution

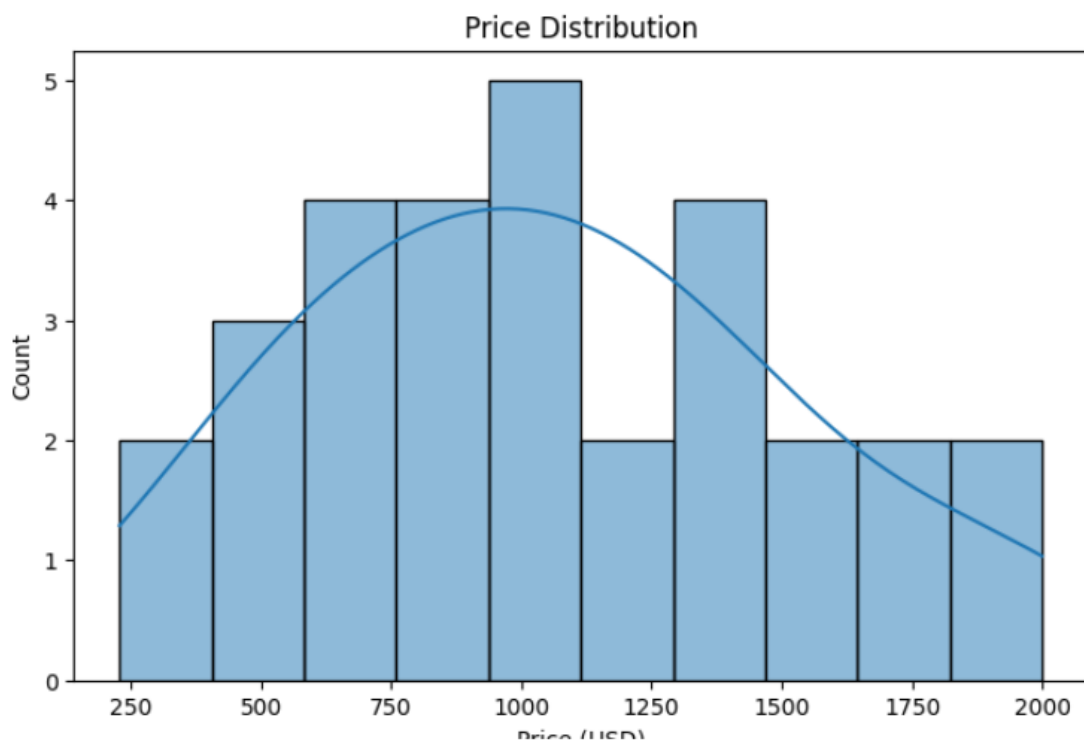
A histogram was used to analyze the distribution of product prices.

Observation: Most products fall within the medium-to-high price range, while a few products belong to the premium segment.

```
plt.figure(figsize=(8,5))

sns.histplot(
    df['Price (USD)'],
    bins=10,
    kde=True
)

plt.title("Price Distribution")
plt.show()
```

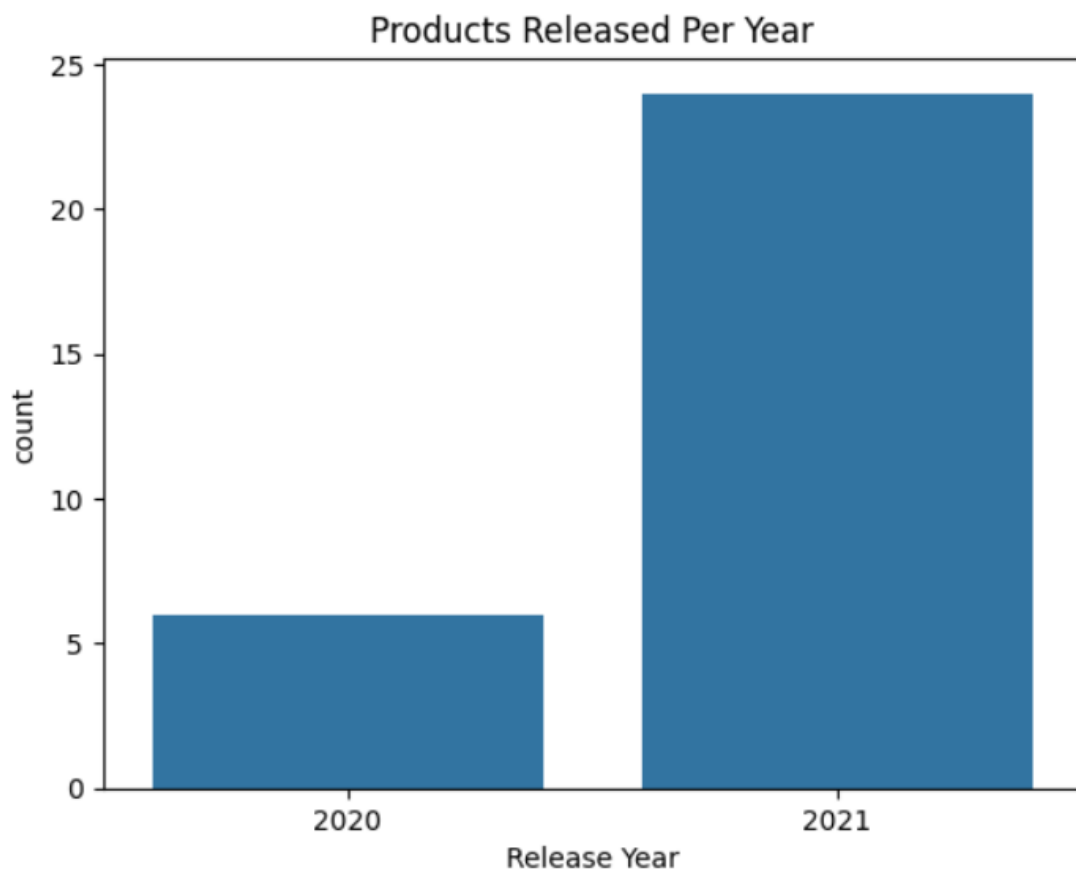


Visualization 5: Release Year Trend

A count plot was generated to analyze product releases by year.

Observation: Most products in the dataset were released during 2021, indicating a focus on current technology products.

```
sns.countplot(  
    x='Release Year',  
    data=df  
)  
  
plt.title("Products Released Per Year")  
plt.show()
```



Key Insights

1. The dataset contains an equal number of products from each brand.
2. The United States dominates the dataset in terms of product representation.
3. Apple products have the highest average pricing among all brands.
4. Product prices vary significantly across different brands.
5. Most products were released in 2021, indicating current market relevance.
6. Premium products contribute to the presence of price outliers in the dataset.

Conclusion

The Technological Products Dataset was successfully cleaned, processed, and analyzed using Python. Data cleaning confirmed that the dataset contained no missing values or duplicate records. Various visualization techniques were used to explore brand distribution, country-wise representation, product pricing, and release trends.

This project enhanced understanding of data preprocessing, exploratory data analysis (EDA), visualization techniques, and data storytelling. The insights obtained from the analysis demonstrate how visualization can help transform raw data into meaningful business information.